

Generative AI as Seniority-Biased Technological Change: Evidence from U.S. Résumé and Job Posting Data*

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Abstract

This paper studies whether generative AI (GenAI) constitutes *seniority-biased technological change*, disproportionately reducing demand for junior workers. We develop a conceptual framework in which GenAI adoption reduces junior labor demand through task displacement and labor-saving productivity gains. We test the framework’s mechanisms and implications using U.S. résumé data covering 65 million workers at more than 280,000 firms, allowing us to track firm-level employment by seniority. GenAI adoption is identified through text analysis that detects ‘GenAI integrator’ job postings, signaling active GenAI implementation by firms. Following adoption, junior employment declines in adopting firms relative to non-adopters, while senior employment trends remain largely unchanged. This decline is concentrated in occupations most exposed to GenAI and is driven primarily by slower hiring rather than increased separations. Moreover, in adopting firms, GenAI-exposed tasks become increasingly less likely to appear in junior task bundles.

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1 Introduction

The impact of generative artificial intelligence (GenAI) on juniors, especially in high-skill, white-collar jobs, has attracted growing attention from both researchers and the media. In many such jobs, workers begin at the bottom of the career ladder performing *intellectually mundane* tasks—routine yet cognitively demanding activities such as debugging code or reviewing legal documents, which are likely to be especially exposed to recent advances in GenAI. As these workers gain experience, they often move up the career ladder to more senior roles that involve more complex problem-solving or managerial responsibilities (Becker, 1964; Garicano, 2000; Ide and Talamas, 2025). If GenAI disproportionately substitutes for entry-level tasks, the lower rungs of these career ladders may be eroding (Garicano and Rayo, 2025).¹

Recent media reports reinforce these concerns (e.g., The New York Times, 2025b; The Wall Street Journal, 2025b). For instance, a July 2025 *Wall Street Journal* article highlighted a sharp drop in demand for junior workers linked to GenAI adoption, drawing on insights from major employers, recruiters, labor market analysts, and recent graduates (The Wall Street Journal, 2025a). One executive at the recruiting firm Hirewell noted that “marketing agency clients have all but stopped requesting entry-level staff—young grads once in high demand but whose work is now a ‘home run’ for AI.”² The issue has also drawn attention from policymakers. At the September 2025 FOMC press conference, Chair Powell was asked about the labor market impact of AI and noted: “You are seeing some effects ... A particular focus on young people coming out of college. Companies may be able to use AI more than they had in the past ... Hard to say how big it is.” (Federal Reserve Board, 2025). Some observers have also linked the diffusion of GenAI to the recent rise in unemployment among college graduates (The New York Times, 2025a; The Atlantic, 2025; Forbes, 2025). Others, however, dispute this interpretation, pointing instead to alternative factors such as macroeconomic uncertainty, post-pandemic adjustment, and increased offshoring (e.g., Financial Times, 2025).

¹The implications can extend beyond short-term employment effects, as early-career earnings strongly influence lifetime income trajectories and, consequently, inequality and mobility (Deming, 2023; Guvenen et al., 2022).

²Moreover, a survey from April 2024 of 804 U.S. hiring managers found that 78% anticipate laying off recent graduates due to AI, with the most vulnerable tasks including research, data entry, email writing, and other routine entry-level assignments (Intelligent, 2024).

This paper aims to measure the potential *seniority-biased* impact of GenAI on the labor market. Specifically, we ask whether GenAI adoption by firms disproportionately affects junior roles relative to more senior positions. This perspective extends the classic literature on skill-biased technological change (e.g., [Katz and Murphy, 1992](#); [Autor et al., 2003](#); [Acemoglu and Autor, 2011](#)), which emphasizes shifts in labor demand across education or occupation groups, to a related but distinct dimension: *seniority*.³

We begin with a simple conceptual framework, building on [Acemoglu and Autor \(2011\)](#) and [Acemoglu and Restrepo \(2018\)](#), to analyze why GenAI adoption may disproportionately reduce demand for junior workers and to clarify the mechanisms underlying this pattern. The framework is motivated by two broad patterns in the emerging evidence: micro-level experimental studies consistently find that GenAI disproportionately increases the productivity of less-experienced and less-skilled workers in a given task ([Brynjolfsson et al., 2025b](#); [Noy and Zhang, 2023](#); [Dell’Acqua et al., 2023](#); [Cui et al., 2025](#); [Cruces et al., 2026](#)), whereas labor market evidence, including our results, links GenAI adoption and exposure to declines in junior employment ([Brynjolfsson et al., 2025a](#); [Azar et al., 2025](#); [Klein Teeselink, 2025](#)). The framework reconciles these patterns.

In the framework, junior and senior workers sort into a continuum of tasks according to comparative advantage. GenAI both automates low-complexity tasks previously performed by juniors and augments productivity on remaining tasks, with larger gains for juniors. The change in junior labor demand decomposes into three forces: a *displacement effect*, capturing the loss of automated tasks; a *productivity effect*, whereby higher productivity reduces labor required per task; and a *reallocation effect*, whereby juniors expand into some previously senior tasks as their relative productivity rises (in the spirit of [Althoff and Reichardt, 2026](#); [Hosseini and Lichtinger, 2026](#)). When tasks are complementary in production, both displacement and productivity effects reduce junior labor demand, while reallocation is the only offsetting force and depends on the extent of differential augmentation. Whether junior employment declines more than senior employment therefore depends on which force dominates.

We then move to our main empirical analysis, drawing on a dataset that combines LinkedIn résumés and job-posting data from Revelio Labs ([Revelio Labs, 2025](#)). The

³See [Ide and Talamas \(2025\)](#) for a closely related theoretical framework, analyzing how GenAI may differentially affect workers within firms depending on their knowledge and role.

dataset covers more than 280,000 U.S. firms, more than 160 million employment spells from roughly 65 million unique workers between 2015 and 2025, and almost 200 million job postings. A key advantage of the data is the standardized seniority classification assigned to each position by Revelio’s algorithm, which enables us to track junior and senior employment within firms over time.⁴

We identify GenAI adoption by detecting job postings that explicitly recruit “GenAI integrator” roles. The method follows the approach of [Hampole et al. \(2025\)](#) and proceeds in two steps: first flagging postings with GenAI-related keywords, then using a large language model (LLM) to determine whether the posting reflects a genuine integrator position—one dedicated to implementing or operating GenAI technology in the firm’s workflow. A firm is classified as an adopter if it has posted at least one such vacancy, thereby capturing firms that have actively initiated the integration of GenAI into their operations.⁵

By this definition, 10,433 firms in our sample adopted GenAI by March 2025. Although adopters represent only about 3.7 percent of the 281,000 firms in our sample, they are substantially larger on average and account for roughly 16 percent of total employment in our dataset. Our analysis shows that adoption of GenAI was minimal and relatively stable prior to 2023, but accelerated sharply thereafter, with a surge of new firms posting integrator roles following the release of ChatGPT in late 2022. We further validate this firm-level adoption measure using worker-side data. Applying a list of GenAI keywords to position descriptions, we find a sharp differential increase in AI-related mentions at adopting firms after the launch of ChatGPT, supporting the interpretation that our adoption measure captures genuine changes in workplace technology use.

The main empirical exercise of the paper is to test the framework’s central implication: whether GenAI adoption disproportionately reduces junior employment relative

⁴We validate the data in Appendices [A.1](#) and [A.2](#). First, we show that the composition of the LinkedIn sample relative to QCEW, OEWS, and QWI benchmarks remains stable over the GenAI diffusion period. Second, we test for differential reclassification: the concern that adopting firms changed how positions were classified as junior after 2022. Using both job-title keywords and workers’ actual labor-market experience, we find that the content of each seniority group evolves similarly at adopting and non-adopting firms, with no break around the ChatGPT launch. This indicates that the decline in junior employment is unlikely to be driven by changes in seniority classification.

⁵Our concept of a GenAI integrator is closely related to the “robot integrator” in [Acemoglu and Restrepo \(2020\)](#), defined as “companies that install, program, and maintain robots.” They use the presence of robot integrators as a proxy for local robot adoption and show that commuting zones more exposed to robots have a higher number of integrators.